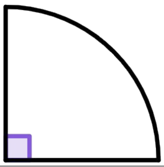
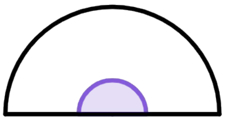
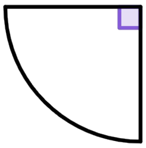
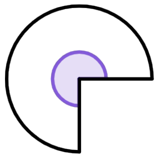
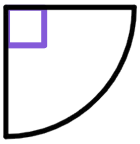
Finding a length in composite shapes

Task A: Finding lengths from areas of circles and sectors

1) Complete the table to find the area of the full circle that the sector comes from.

				
area of sector	10π cm²	25.1 cm²	31π cm²	60π cm²
calculation	$10\pi \times 4$	25.1×2		
area of full circle	40π cm²			

2) The area of this quarter-circle is 900π cm².



a) Find the area of the full circle that the sector comes from.

b) Using your answer to part a, complete the method to calculate the radius of this quarter-circle.

area of full circle = $\pi \times r^2$

$\boxed{} = \pi \times r^2$

$\boxed{} = r^2$

60 cm = r

3) The area of a semicircle is 39.3 cm².

Complete the method to calculate the diameter of this semicircle.

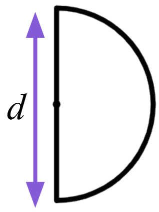
area of full circle = $\pi \times r^2$

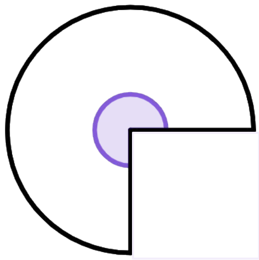
$\boxed{} = \pi \times r^2$

$\boxed{} = r^2$

$\boxed{} = r$

$\boxed{} = d$



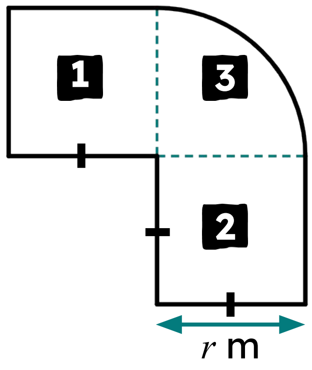


4) The area of this three-quarter-circle is 1357.2 cm².

What is the radius of the sector?

Task B: Finding lengths of composite shapes using algebra

1) This composite shape has an area of 470.7 m².



a) Write down the areas, in terms of r, for the three component parts of this shape.

area of **1**

area of **2**

area of **3**

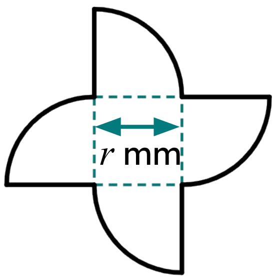
b) Show by factorising that the area of the composite shape can be written as:

$$r^2 \times 2.785\ldots$$

c) Complete the calculation to find the radius of the quarter-circle, given to the nearest integer.

$$r^2 \times 2.785\ldots = 470.7$$

2) This composite shape has an area of 11 633.7 mm².
Complete the steps to calculate the value of r, giving your answer to the closest integer.



area: $\boxed{}$ (circle) + $\boxed{}$ (square) $= 11633.7$

factorise: $r^2 \boxed{()} = 11633.7$

convert π expression into decimal: $r^2 \boxed{()} = 11633.7$

divide: $r^2 = \boxed{}$

square root: $r \approx \boxed{}$

3) This composite shape has an area of 4031 cm².
Find the value of r, giving your answer to the closest integer.